

pET28a-pT5-IF2

Created: 4/10/2025, 10:28:41 AM

Saved: 4/10/2025, 10:53:55 AM

Component Fragments

Name	Length	Produced by	5' End	3' End
pET28a-ccdB	5143	PCR	Fwd Primer (auto)	Rev Primer (auto)
pT5-IF2	2834	PCR	Fwd Primer (auto)	Rev Primer (auto)



Notes

- Primer pET28a-ccdB_rev contains a run of 4+ repeats of a mono/di/trinucleotide.
- Primer pT5-IF2_fwd contains a run of 4+ repeats of a mono/di/trinucleotide.
- Primer pT5-IF2_fwd has %GC outside of desired range (35-65%) in the annealing segment.

Required oligos

Name	Primer 5' (overlap/spacer/ANNEAL) 3'	Len	%GC	3' %GC	3' Tm	3' Ta
pET28a-ccdB_fwd	cattgcttaaGCTTGATCCGGCTGCTAAC	29	48	58	66.4	65.6
pET28a-ccdB_rev	tttttatgaATTTTCGCGGGATCGAGATC	29	38	53	64.6	65.6
pT5-IF2_fwd	cccgcgaaatTCATAAAAAATTTATTTGCTTTGTG	35	31	20	55.7	56.7
pT5-IF2_rev	cggatcaagcTTAAGCAATGGTACGTTG	28	46	39	56.5	56.7

Build Settings

Property	Value
Product/Kit	#E5520 NEBuilder HiFi DNA Assembly Cloning Kit
Minimum Overlap	20 nt
Minimum Overlap Tm	48 °C
Circularize	Yes
PCR Polymerase/Kit	Q5 High-Fidelity DNA Polymerase
PCR Primer Conc.	500 nM
Min. Primer Length	18 nt

Assembled Sequence

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#ACCESSION   .
#VERSION     .
#KEYWORDS    NEBuilder
#SOURCE      synthetic DNA construct
# ORGANISM   synthetic DNA construct
#REFERENCE   1 (bases 1 to 7937)
# AUTHORS    .
# TITLE      NEBuilder-generated Construct
# JOURNAL     Exported 10-APR-2025 from NEBuilder https://nebuilder.neb.com
#COMMENT     NEBuilder-generated oligos (UPPERCASE = gene-specific, lowercase = overlap)
#COMMENT     pET28a-ccdB_fwd: cattgcttaaGCTTGATCCGGCTGCTAAC
#COMMENT     pET28a-ccdB_fwd 3'Tm: 66.4 3'Ta: 65.6
#COMMENT     pET28a-ccdB_rev: ttttttatgaATTTCGCGGGATCGAGATC
#COMMENT     pET28a-ccdB_rev 3'Tm: 64.6 3'Ta: 65.6
#COMMENT     pT5-IF2_fwd: cccgcgaaatTCATAAAAAATTATTGCTTTGTG
#COMMENT     pT5-IF2_fwd 3'Tm: 55.7 3'Ta: 56.7
#COMMENT     pT5-IF2_rev: cggatcaagcTTAAGCAATGGTACGTTG
#COMMENT     pT5-IF2_rev 3'Tm: 56.5 3'Ta: 56.7
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#                     /note="overlap: cggatcaagc"
#                     /note="gene-specific primer: TTAAGCAATGGTACGTTG"
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